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pBRICS: A Novel Fragmentation Method for Explainable Property Prediction of Drug-Like Small Molecules

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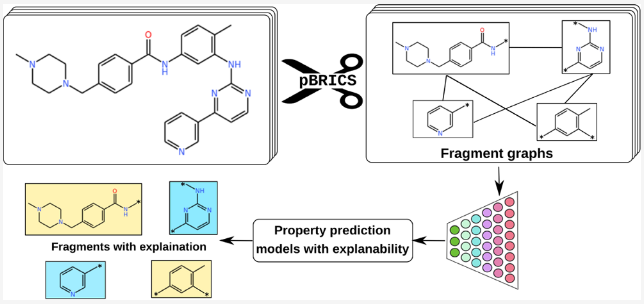

ABSTRACT: Generative artificial intelligence algorithms have shown to be successful in exploring large chemical spaces and designing novel and diverse molecules. There has been considerable interest in developing predictive models using artificial intelligence for drug-like properties, which can potentially reduce the late-stage attrition of drug candidates or predict the properties of novel AI-designed molecules. Concurrently, it is important to understand the contribution of functional groups toward these properties and modify them to obtain property-optimized lead compounds. As a result, there is an increasing interest in the development of explainable property prediction models. However, current explainable approaches are mostly atom-based, where, often, only a fraction of a fragment is shown to be significant. To address the above challenges, we have developed a novel domain-aware molecular fragmentation approach termed post-processing of BRICS (pBRICS), which can fragment small molecules into their functional groups. Multitask models were developed to predict various properties, including the absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties. The fragment importance was explained using the gradient-weighted class activation mapping (Grad-CAM) approach. The method was validated on data sets of experimentally available matched molecular pairs (MMPs). The explanations from the model can be useful for medicinal chemists to identify the fragments responsible for poor drug-like properties and optimize the molecule. The explainability approach was also used to identify the reason behind false positive and false negative MMP predictions. Based on evidence from the existing literature and our analysis, some of these mispredictions were justified. We propose that the quantity, quality, and diversity of the training data will improve the accuracy of property prediction algorithms for novel molecules.

■ INTRODUCTION

The advent of artificial intelligence (AI)-based algorithms has revolutionized the complete drug design process. There has been significant progress in the field of generative AI to design novel drug-like molecules by effectively exploring the chemical space.[1−10] Some studies have shown the potential of these models to truly accelerate the early stage drug discovery process and design drug-like molecules.[11,12] While it is essential to explore the potentially vast chemical space for the generation of novel drug-like small molecules specific to a target protein of interest, it is also necessary to optimize various physicochemical and late-stage drug-like properties including absorption, distribution, metabolism, excretion, and toxicity (ADMET) to reduce the late-stage attrition of novel drug molecules.[8,13] The ability to accurately predict various drug-like properties can provide insights right from the administration of a molecule to its elimination, along with its probable pharmacological action in the human body.[14−16] Studies have shown that around 50% of potential drug candidates fail while being evaluated for their pharmacokinetic properties, leading to a marked increase in the time taken to produce admissible drug molecules.[17,18] To address this, several machine learning-based models were proposed to predict target-specificity, physicochemical, and ADMET properties.[19−27] It was proposed that the inclusion of these properties during the early stages of drug design can effectively reduce late-stage attrition.[8] One of the challenges for such predictive models is to differentiate between highly similar molecules that can be categorized into different classes depending on their property profile. Consequently, there has been considerable interest in validating various machine learning and deep learning models by assessing their performance in classifying similar molecules with very different properties (also known as activity cliffs).[28,29]

Apart from measuring the performance of similar molecules, the interpretability of model predictions can help during the lead

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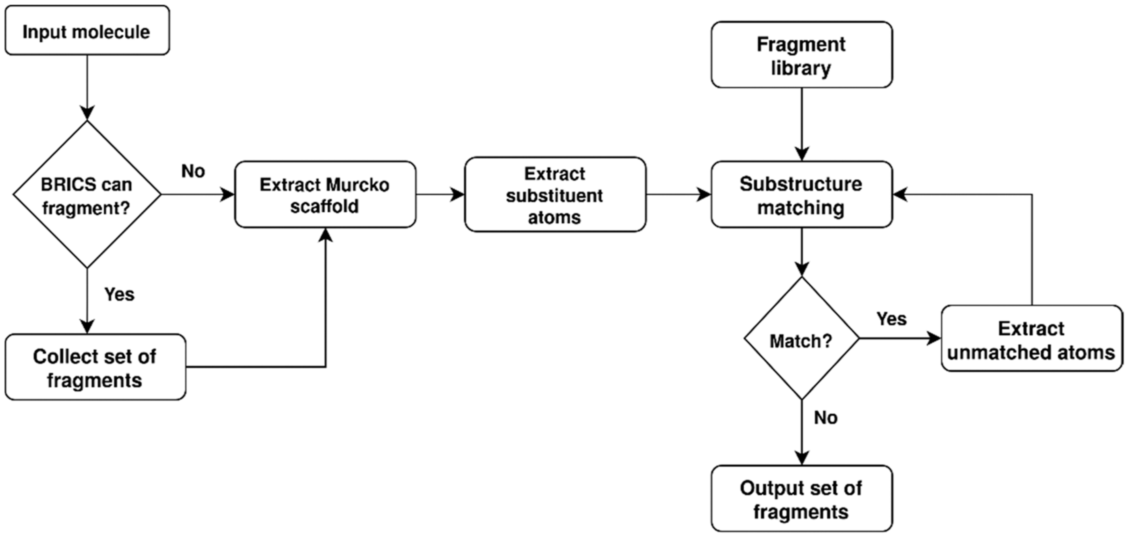

Figure 1. Flowchart of molecule fragmentation using the pBRICS method.

optimization.[26,30−32] The ability to quantify the contribution of each substructure or molecular fragment toward the property being predicted can guide medicinal chemists to optimize the lead molecules. The choice of input representation of molecules can influence what the model learns and, consequently, the explanations obtained for model predictions. A few of the widely used input representations for the development of predictive models include the simplified molecular input line entry system (SMILES), atomic-level molecular graphs, fingerprints based on functional group frequency, and descriptor-based feature vectors such as PaDEL[33] and Mordred.[34] For models developed using each of the above input representations, corresponding model explanations have also been extracted using gradient-based explainability methods such as gradient-based class activation mapping (Grad-CAM)[35] and saliency maps and nongradient explainability methods such as SHAPley scores (SHAP)[36] and local interpretable model-agnostic explanations (LIME).[37] The explanations from molecular graph representations typically involve coloring schemes to highlight the magnitude of gradient change. However, these methods often produce partial explanations for functional groups. For example, only a few atoms of ring systems and functional groups are highlighted as important.[38] To remedy this, functional group level molecular graph representation could be a solution where the functional groups are nodes and fragmented bonds as edges.[39−41] However, there are limitations with most of the existing fragmentation methods such as breaking of retrosynthetically interesting chemical substructures (BRICS), REtrosynthetic combinatorial analysis procedure (RECAP), and synthetic disconnection rules[42] (SynDiR), which lead to singleton atoms upon fragmentation, cannot handle very small molecules, and fail to fragment all substituents of a core scaffold in a molecule. Specifically, some substituents, including halogen atoms and methyl and hydroxyl groups, can be considered valid fragments of interest to medicinal chemists due to the possibility of extensive change in the physicochemical and ADMET properties of a molecule upon their substitution.[43−45] The above limitations necessitate the development of a novel fragmentation method that could address the above challenges and could be used as an input to train deep learning models by capturing a chemist's perception.

In this study, we have addressed the challenges involved in quantifying the fragment-level contributions toward property prediction using a gradient-based explainability method with the help of a novel fragment-based molecular graph representation. A novel molecular fragmentation method termed pBRICS was proposed for the fine-grained fragmentation of any novel small molecule. The molecular graph obtained based on the fragments of pBRICS was used to train graph convolution network (GCN) models for the prediction of over 40 different ADMET properties. The models developed were compared with an existing atom-level graph model from the literature trained using a similar architecture. The explanations obtained from the predictive models were validated based on their relevance to the existing knowledge base of substructure contributions using matched molecular pair (MMP) analysis.

■ MATERIALS AND METHODS

Fragmentation of Molecules Using the Post-Processing BRICS (pBRICS) Method. In this work, we have introduced a novel fragmentation method named pBRICS. The pBRICS method performs further fragmentation of the results obtained from the commonly used BRICS method.[46] The BRICS method considers 16 chemical environments and their corresponding fragment prototypes for the fragmentation of every bond in a molecule. Due to the expanded set of cleavage rules possible for a molecule, the BRICS method[46] can fragment 10% more molecules than the RECAP fragmentation method.[47] The pBRICS method attempts to iteratively fragment the molecule such that the smallest possible substituents of a scaffold are also enumerated, which can be of interest in building a knowledge base for molecular optimization problems.

pBRICS uses a comprehensive library of fragments (ring systems and nonring substituents) manually curated from the Open Chemistry project database (https://github.com/OpenChemistry) and the literature. If a molecule can be fragmented by the BRICS method, the resultant fragments are primarily classified into scaffolds and substituents. The scaffold of a molecule or a fragment is defined based on the Bemis−Murcko framework[48] as implemented in RDKit. The atoms corresponding to substituents are further matched against the library of manually curated fragments to obtain the largest possible fragment match, and the process is iterated until no

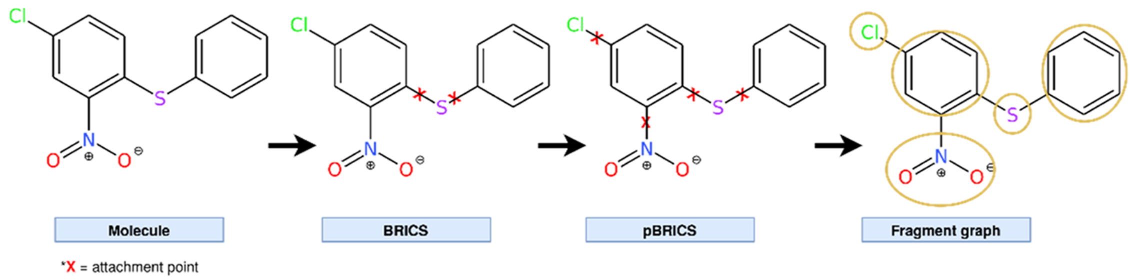

Figure 2. Graph construction process starting from the molecule of interest and subsequent BRICS, pBRICS fragmentation, and final fragment graph formation. The post-processing of BRICS fragments allowed the pBRICS method to further divide the molecules into smaller and chemically interesting fragments.

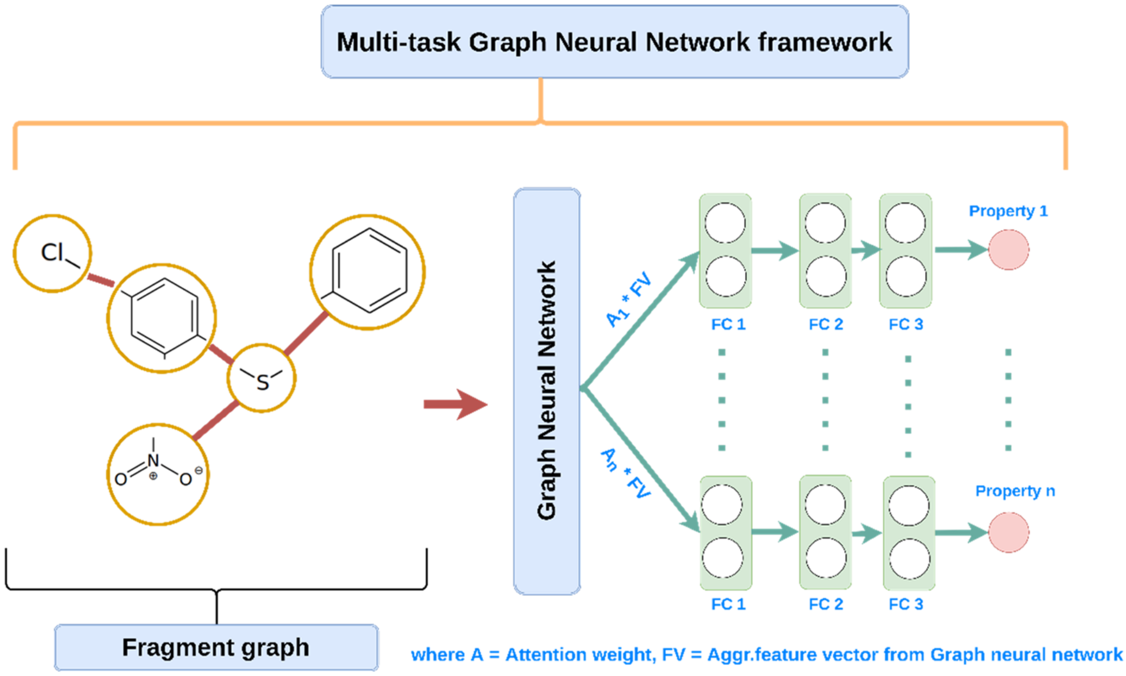

Figure 3. Multitask graph neural network framework proposed in this study for the prediction of 40 pharmacokinetic properties. The molecules in the graph format are provided as an input to the mulltitask GCN model. The resulting aggregated feature vector from the graph model is then used to predict the pharmacokinetic properties through independent, fully connected layers. *n* corresponds to the number of properties being considered.

further fragment matches are possible. If a molecule cannot be fragmented by the BRICS method, the atoms corresponding to the scaffold and substituents of the molecule are enumerated and the iterative fragmentation procedure is applied to the substituent atoms instead of fragments from the BRICS method (Figure 1). The pBRICS method was implemented in Python using the utilities from the RDKit library.

Graph Construction from Fragments Obtained from the pBRICS Method. The fragments obtained from pBRICS were used to construct a domain-aware fragment-level graph representation. For generating the graph, fragments were considered as nodes and an edge was defined between two nodes if the two fragments were connected in the molecule by at least one bond (see, Figure 2).

Once the graph construction is complete, the next step is to define the feature vector for each node in the graph. The MACCS (Molecular ACCess System) 167 keys'[49] representation of fragment (node) is considered as the feature vector for each node in the graph. The type of bond connecting the fragments is encoded as a value to be used as edge weight while aggregating the information from neighbors.

Data Set Curation. The proposed method was implemented to predict 40 pharmacokinetic properties. The corresponding data were curated from the therapeutic data commons (TDC)[50] [https://tdcommons.ai/] repository. For each property, the data were split into 80% for the training set, 10% for the validation set, and the remaining 10% for the test set. The individual data sets for these 40 properties were further combined to form a multitask data set to be used for training a multitask property prediction model. Out of 40 properties, 30 properties belong to the classification category and 10 belong to the regression category.

Graph Convolutional Network (GCN). The domain-aware graph generated using the pBRICS fragmentation method was used as an input to the deep learning models involving graph convolutional network (GCN)[51] layers as implemented in the Deep Graph Library.[52] The GCN-based architecture has been proven to yield promising results for predicting a wide range of molecular characteristics[53] Typically, a graph is defined as $G = (V,E)$, wherein fragments are nodes (V) and bonds connecting them are edges (E). The GCN layer modifies each node's embedding in the following way: (1) Using the node feature matrix $X \in F^{n\times d}$ and an adjacency matrix $A \in \{0,1\}^{n\times n}$, GCN aggregates the information from neighboring nodes.[54] (2) Applies a nonlinear activation function to the aggregated feature vector embedding of nodes[54] to get the prediction.

Construction of a Multitask Graph Neural Network Framework. The multitask graph neural network model (Figure 3) was applied so that the model can benefit from related tasks and improve the prediction for tasks with low data. The multitask model can be used for predicting both classification and regression properties simultaneously, unlike the single-task predictive models, where individual models must be built independently for each task (property) of interest. Recent studies have shown that multitask models can outperform single-task models,[27,55] as the hidden layers are shared among all of the tasks, thereby helping the model to learn more about the related tasks, which will improve the performance of the task at hand. In the multitask model, a batch of fragment graphs is first passed into a two-layered GCN each of size 64. The aggregated feature vector (FV) from the final GCN layer for each graph is multiplied by their respective attention weight, and the resultant vector is fed into an independent stack of fully connected (FC) layers (one FC stack per property). Here, all FC stacks share an aggregated feature vector from the final GCN layer, enabling the model to learn common features between all combinations of properties involved.

Evaluation Metrics. The binary cross-entropy-loss (BCE) (eq 1) between the actual values and model predictions is calculated for each of the 30 classification properties and mean squared loss (MSE) (eq 2) for the 10 regression properties. The combined loss is back-propagated to update the weights of the neurons in the model. The area under receiver operating system curve (auROC) was used for measuring the performance of the classification models and the Spearman correlation was used for measuring the performance of regression models in this study.

$$\mathrm{BCE} = \frac{-1}{N}\sum_{i=1}^{N}(y_i \log(\overline{y_i}) + (1 - y_i)\log(1 - \overline{y_i})) \tag{1}$$

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \overline{y_i})^2 \tag{2}$$

Explainability of Model Predictions Using the Gradient Class Activation Map (Grad-CAM). To interpret the predictions obtained from the deep learning model, the gradient class activation map (Grad-CAM) method was used. The Grad-CAM method was initially developed for analyzing convolutional neural network (CNN) models and was later adapted to work with graph convolutional networks (GCN).[3] Here, consider c to be the class for which explanations are generated, L to be the final GCN layer of the model, $\alpha_k^{L,c}$ (eq 3) to be the Grad-CAM weight for the kth feature of class c at layer L, N to be the number of nodes, and $F_{k,n}^{L}$ to be the feature map of the kth feature map at layer L of node n:

$$\alpha_k^{L,c} = \frac{1}{N}\sum_{n=1}^{N}\frac{\partial y^c}{\partial F_{k,n}^{L}} \tag{3}$$

Node level importance scores for a graph obtained using the Grad-CAM method are given by the following equation (eq 4). Based on these scores, one can identify the fragments (nodes) contributing positively/negatively to the class c.

$$L_{\mathrm{Grad-CAM}}^{c}[L, n] = \sum_{k}\alpha_k^{L,c}.\ F_{k,n}^{L}(X, A) \tag{4}$$

Validation of the pBRICS Method with a Matched Molecular Pair (MMP). One of the major challenges for the predictive model is to correctly predict similar molecules that belong to conflicting classes in a classification problem. Experimentally validated MMPs[56] were considered that differ by a single substitution of functional group/atoms. MMPs provide an opportunity to validate such models by testing their ability to differentiate similar molecules belonging to opposite classes. Further, by extracting the corresponding reasoning from explainable AI methods, model predictions can also be justified. An example of an MMP is shown in Supporting Information (Figure S1).

RESULTS AND DISCUSSION

Benchmarking pBRICS against Other Fragmentation Methods. The pBRICS fragmentation approach proposed in this study was compared with the existing fragmentation methods, namely, BRICS, RECAP, and SynDiR, in terms of four metrics defined earlier[42] with the ChEMBL data set as the benchmark data set (Table 1). Based on the comparison results,

Table 1. Comparison of the Performance Metrics between Different Fragmentation Methods on the ChEMBL Database

metric	pBRICS	BRICS	RECAP	SynDiR
no. of unfragmented molecules	5908	55 677	152 177	97 823
total no. of fragments generated (with duplicates)	11 913 619	8 328 962	4 964 387	5 672 031
no. of unique fragments	120 552	242 208	242 208	342 263
average heavy atom count for the unique fragments	12.201	15.593	15.911	17.693

it is notable that pBRICS can fragment 16-, 5-, and 10-times more molecules than RECAP, BRICS, and SynDiR methods, respectively. However, pBRICS generates the least number of unique fragments with the smallest heavy atom count, resulting in a more generic fragment library due to the fine-grained fragmentation procedure developed. Several examples were identified, wherein only pBRICS could fragment the molecule of interest (Figure S2).

Performance of the Multitask Property Prediction Model. As mentioned in the Materials and Methods section, the molecules represented using domain-aware molecular graph representation based on the pBRICS were used to train multitask models for predicting 40 properties that are crucial for small molecules to be considered drug-like molecules. Among these 40 properties, 30 properties are based on the classification task and 10 properties are based on the regression task. A single GCN-based multitask model, referred to as MT-Frag, was trained on all 40 properties. Next, two separate multitask models were trained to handle classification and regression tasks, respectively, to check if they can improve performance. The resultant model was labeled MT-Frag-Mod. The main difference between MT-Frag and MT-Frag-Mod is that in the earlier, both classification and regression tasks were trained together. MT-Frag and MT-Frag-Mod were compared with an atom-based multitask graph attention framework (similar to the model proposed by ADMETLab2.0[27]) and this model is referred to as MT-Atom in Tables 2 and 3. MT-Atom

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Table 2. Performance (AuROC Score) of Proposed Multitask Models (MT-Frag and MT-Frag-mod) and their Comparison with the MT-Atom (Based on ADMETLab2.0) Model in 30 Classification Data Sets[a,b]

property	data set size	MT-frag	MT-frag-mod	MT-atom
HIA	578	**0. 985**	0.896	0.956
Pgp	1212	0.924	**0.930**	0.928
bioavailability	640	0.514	**0.642**	0.601
BBBP	1975	0.881	0.840	**0.918**
CYP2C19 inhibition	12 665	0.853	**0.865**	0.862
CYP2D6 inhibition	13 130	0.785	**0.843**	0.826
CYP3A4 inhibition	12 328	0.800	**0.850**	0.832
CYP1A2 inhibition	12 579	0.853	**0.889**	0.884
CYP2C9 inhibition	12 092	0.836	**0.870**	0.859
CYP2C9 substrate	666	0.755	**0.812**	0.756
CYP2D6 substrate	664	**0.805**	0.802	0.771
CYP3A4 substrate	667	0.616	**0.633**	0.598
NR-AR	7831	0.752	0.792	**0.805**
NR-AR-LBD	7831	0.788	0.884	**0.914**
NR-AhR	7831	0.880	0.875	**0.882**
NR-aromatase	7831	0.764	**0.880**	0.802
NR-ER	7831	0.681	**0.706**	0.701
NR-ER-LBD	7831	0.715	**0.763**	0.709
NR-PPAR-γ	7831	0.757	0.852	**0.917**
SR-ARE	7831	0.745	**0.827**	0.789
SR-ATAD5	7831	0.802	**0.897**	0.865
SR-HSE	7831	0.692	**0.781**	0.745
SR-MMP	7831	0.867	0.897	**0.903**
SR-p53	7831	0.816	**0.853**	0.837
hERG	7831	0.913	0.826	0.809
Ames mutagenicity	7255	0.822	**0.824**	0.799
DILI	475	0.817	0.832	**0.897**
skin reaction	404	**0.723**	0.629	0.689
carcinogens	278	0.825	0.901	**0.906**
clinTox	1484	**0.846**	0.782	0.705
mean performance		0.797	**0.822**	0.813

[a]The data set size includes train, validation, and test data. [b]HIA—human intestinal absorption; Pgp—*P*-glycoprotein; BBBP—blood–brain barrier permeability; CYP2C19—CYtochrome P450 family 2 subfamily C member 19; CYP2D6—CYtochrome P450 family 2 subfamily D member 6; CYP2A4—CYtochrome P450 family 2 subfamily A member 4; CYP1A2—CYtochrome P450 family 1 subfamiy A member 2; CYP2C9—CYtochrome p450 family 2 subfamilly C member 9; NR-AR—nuclear receptor androgen receptor; AhR—aryl hydrocarbon receptor; ER—estrogen receptor; PPAR-γ—peroxisome proliferator-activated-γ; SR-ARE—stress response-antioxident response element; HSE—heat shock factor response element; p53—signaling pathways; MMP—mitochondrial membrane potential; hERG—human ether-a-go-go related gene; DILI—drug induced liver injury; and ClinTox—clinical toxicity.

from ADMETLab2.0 was retrained with the same training and test data sets used for MT-Frag and MT-Frag-Mod for fair comparison. The performance of each multitask model in classification properties is summarized in Table 2 and regression properties is summarized in Table 3. The hyperparameters utilized in this study have been included in the Supporting Information (Table S1).

The performance of the proposed fragment-based multitask models (MT-Frag and MT-Frag-Mod) was compared with the atom-level model from ADMETLab2.0 (MT-Atom). On classification properties, MT-Frag-Mod achieved a mean performance of 82.2% after hyperparameter tuning, which is

Table 3. Performance (Spearman Correlation) of the Proposed Multitask Models (MT-Frag and MT-Frag-mod) and their Comparison with the MT-Atom (Based on ADMETLab2.0) Model in 10 Regression Data Sets[a,b]

property	data set size	MT-frag	MT-frag-mod	MT-atom
Caco2	906	0.767	**0.814**	0.775
lipophilicity	4200	0.701	0.741	**0.759**
solubility	9982	0.771	0.826	**0.842**
hydration free energy	642	0.813	**0.875**	0.859
PPBR	1614	0.587	**0.682**	0.588
VDss	1130	0.479	0.338	**0.525**
half-life	667	0.348	**0.452**	0.244
clearance_hepatocyte	1102	0.444	**0.501**	0.449
clearance_microsome	1120	0.523	**0.569**	0.541
LD50	7385	0.623	0.619	**0.670**
mean performance		0.605	**0.641**	0.625

[a]The data set size includes train, validation, and test data. [b]Caco2—cancer coli 2 cell effective permeability; PPBR—plasma protein binding rate; VDss—volume of distribution at steady state; and LD50—median lethal dose.

1% higher than that of the MT-Atom model (81.3%) and 2.4% higher than that of the MT-Frag model (79.7%). In regression properties, MT-Frag-Mod achieved a mean performance of 64.1%, which is 1.5% higher than that of the MT-Atom model (62.5%) and 4% higher than that of the MT-Frag model (60.5%). The increase in performance of MT-Frag-Mod can be attributed to the training of classification and regression properties separately. For the 40 properties considered in the study, it was observed that MT-Frag-Mod achieved top performance in 24 properties, whereas MT-Frag achieved top performance in only 4 properties. On the other hand, the MT-Atom model achieved top performance in 12 out of 40 properties. To determine if there is a significant difference between the three models (MT-Frag, MT-Frag-Mod and MT-Atom), a *t*-test was performed at a significance level of 0.05. The mean performance of MT-Frag-Mod and MT-Atom is comparable and different from MT-Frag (Tables 2 and 3). This is reflected in the t-test, where both MT-Frag-mod and MT-Atom models performed significantly better than the MT-Frag model (Table S2).

In addition to this, we have conducted a comparison study between pBRICS and models employing other fragmentation methods such as BRICS, RECAP, and SynDiR. The corresponding results for multitask models using BRICS, RECAP, and SynDiR (MT-BRICS, MT-RECAP, and MT-SynDiR) are shown in the Supporting Information (Tables S3 and S4, corresponding to multitask classification tasks and multitask regression tasks, respectively). Overall, the method proposed in this work performed better, which can be observed from the mean performance of the tasks.

Overall, MT-Frag-Mod was observed to be a better predictive model in the multitask scenario. Once the performance of the models was established, the fragment-level explanations to model predictions were extracted. As explained in the method section, data sets of matched molecular pairs (MMPs) were curated for each property, wherein the most number of MMPs were obtained for BBB permeability and Ames mutagenicity.

Matched Molecular Pair (MMP) Analysis for BBB Permeability (BBBP). One of the major challenges for the predictive model is to accurately predict small molecules from

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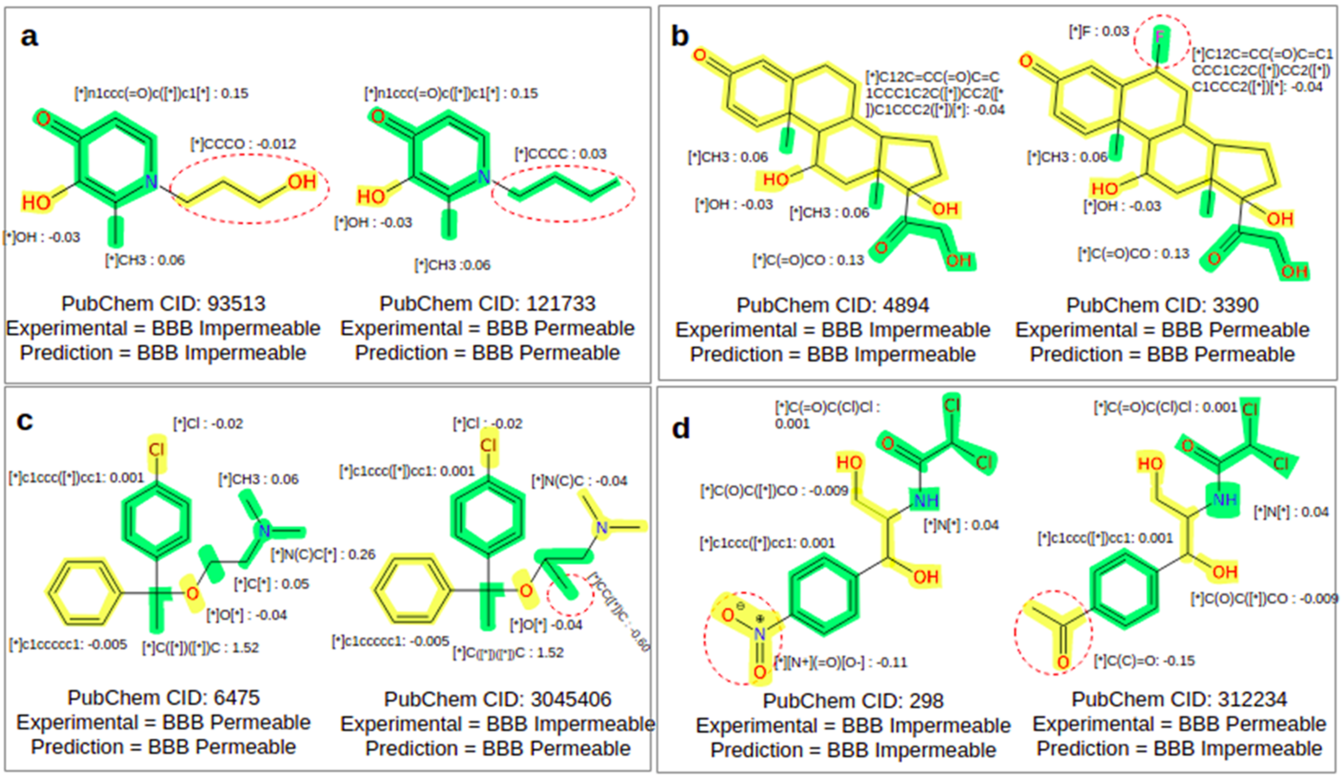

Figure 4. Examples of correctly predicted MMPs (a, b) and incorrectly predicted MMPs (c, d) using the MT-Frag-Mod model for the BBBP property. The MMP transformation is highlighted in the dotted red circle. The fragments contributing to BBB permeable and BBB impermeable molecules are represented by the green and yellow colors, respectively. The average Grad-CAM value of fragments in the correctly predicted samples is shown next to the respective fragment.

the MMP data set (pairs that differ marginally by a single functional group change but exhibit contrasting properties) such that their explanations correlate well with the model prediction. The MMP data set for BBBP property consists of 102 unique pairs of molecules along with their corresponding ground-truth labels. To check the model performance on the MMP data set, the molecules from all pairs of MMP entries were combined to form a unique set of 110 molecules. The MMPs identified were used to compare the two trained models, namely, MT-Frag-Mod and MT-Atom (based on ADMETLab2.0), and an external model, SwissADME.[22] The proposed method was found to accurately predict 37.3% of the molecules in the MMP data set, surpassing the ADMETLab2.0's prediction rate of 30% by 7.3% and exceeding the swissADME's prediction rate of 31.8% by 5.5%. The corresponding results are summarized in Supporting Information, Table S5. While there is an increase in the prediction of MMPs using the current method when compared to existing methods, it should be noted that the prediction of MMPs correctly is a challenging task, as also discussed in previous works.[28]

To obtain the explanations, the Grad-CAM algorithm was applied to all of the molecules in the MMP data set to get the gradient weights corresponding to each node (fragment contribution). First, the contribution of each fragment of all of the molecules is calculated according to the model prediction using the Grad-CAM approach. Next, the Grad-CAM value for all of the unique fragments is collected from all of the correctly classified positive molecules (molecules for which predictions and experimental result matches) and averaged. The averaging resolved the issue of variability around the single Grad-CAM value and helped in deriving suitable decisions on MMP pairs. The molecules are colored according to the average Grad-CAM values of each unique fragment (Figure 4). It is important to note that the addition of Grad-CAM values of all individual fragments does not result in the total contribution of molecules with or without averaging.

In the case of BBB permeability, BBB permeable and impermeable fragments are part of the positive and negative classes, respectively. The fragments from positive and negative classes are shown in green and yellow, respectively (Figure 4). Figure 4 depicts two examples of correctly predicted MMP pairs from the data set, whose Grad-CAM values were found to match with their labels (experimentally verified BBB permeable or impermeable). In the MMP pair shown in Figure 4a, replacing the methyl group with a hydroxyl group transformed the molecule from BBB permeable to BBB impermeable. In the case of Figure 4b, replacing the halogen (fluorine) with a hydrogen atom transformed the molecules from BBB permeable to BBB impermeable. It is notable that these transformations are also experimentally proven to disrupt the BBBP property of a molecule.[43] For example, usually, the presence of hydrophobic and halogen groups increase BBB permeability, and the presence of polar and charged residues reduces the chances of BBB permeability of small molecules.[43,45] On similar lines, a positive fragment contribution (in terms of the Grad-CAM gradient) of the methyl group was observed to be contributing toward BBB permeability. Similarly, the halogen group increased the BBB permeability for the molecule in Figure 4b. Bung et al.[8] verified that the presence of a hydroxyl group in the designed molecule (a highly similar novel molecule to Robalzotan, which is known to be a potent BBB permeable lead compound targeting the 5-HT1A receptor) increased its polarity and resulted in a low

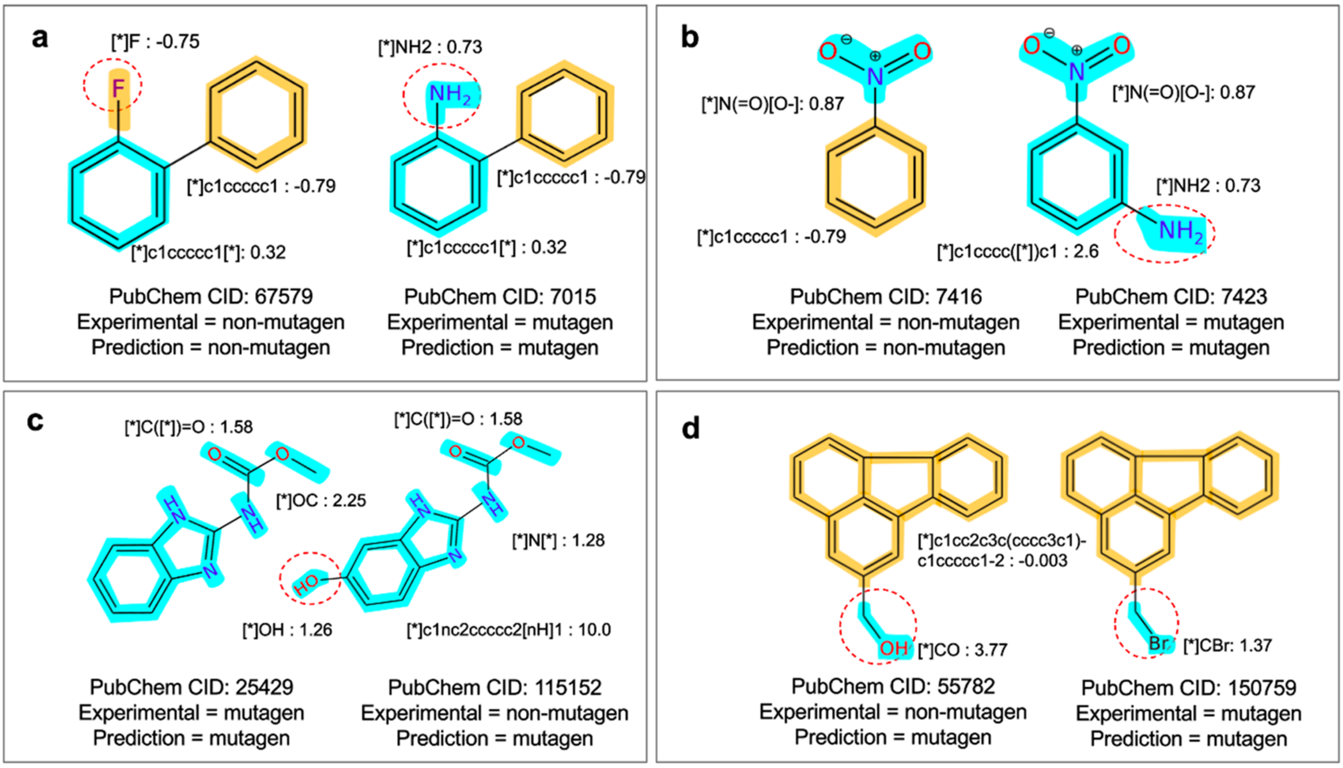

Figure 5. Examples of correctly predicted MMPs (a, b) and incorrectly predicted MMPs (c, d) using the MT-Frag-Mod model for the Ames mutagenicity property. The MMP transformation is highlighted in the dotted red circle. The fragments contributing to Ames mutagenicity and Ames nonmutagenicity are represented by the cyan and orange colors, respectively. The average Grad-CAM value of fragments in the correctly predicted samples is shown next to the respective fragments.

predicted BBBP probability.[8] To understand the reason behind the false positives and false negatives from the Grad-CAM analysis, incorrectly predicted MMPs were analyzed to justify the rationale behind the model prediction. Two such examples are shown in Figure 4c,d.

In Figure 4c, although the second molecule is an experimentally proven BBB impermeable molecule, the model predicted it to be BBB permeable. It is well known that the hydrophobic fragments (here, hydrogen to methyl group transformation) are susceptible to BBB permeability. As discussed before, the average Grad-CAM of the methyl group was found to be higher in the set of BBB permeable molecules. As a result, the model has interpreted it as a BBB permeable molecule. Provided the left molecule (Pubmed CID 6475) in Figure 4c is an experimentally verified BBB permeable molecule (which is in agreement with MT-Frag-Mod), it is unlikely that the methyl group substitution would make the MMP molecule (Pubmed CID 3045406) BBB impermeable. Along similar lines, although the second molecule (Pubmed CID: 312234) in Figure 4d is experimentally verified to be BBB permeable, the model predicted it as a BBB impermeable molecule. The Grad-CAM analysis showed that the fragment responsible for the MMP transformation got a negative Grad-CAM score and is frequent in the BBB impermeable molecules. As a result, the model has predicted the corresponding molecule as BBB impermeable. Based on the fragment contribution and evidence available from the existing literature on BBB permeability,[43,45] it may be possible that the predictions for molecules shown in Figure 4c,d are correct but experimental validation is required to prove this further. The explainable fragment approach proposed in this work can be a valuable tool to understand the reason behind any predictions from the AI methods. The availability of an explainable fragment method can also identify the fragment responsible for a poor drug-like property and provide an opportunity to convert hit to lead molecules through a better in silico approach.

Matched Molecular Pair (MMP) Analysis for Ames Mutagenicity. A similar analysis of the MMP data set was performed for Ames mutagenicity. The Ames mutagenicity data set consists of 600 unique pairs of molecules along with their corresponding ground-truth labels (experimental results). To check the model performance on the MMP data set, the molecules from all pairs of MMP entries for Ames mutagenicity were combined to form a unique set of 712 molecules. The MMPs identified were supplied to both trained models (MT-Frag-Mod and MT-Atom) and one external model (swissADME) to compare their performance (Table S6). The proposed method was found to accurately predict 55.5% of the molecules in the MMP data set, surpassing the MT-Atom's (based on ADMETLab2.0) prediction rate of 52.1% by 3.3% and exceeding the swissADME's prediction rate of 47.5% by 8% (Table S6).

Based on Grad-CAM analysis, replacing an amino group with the fluorine atom or hydrogen atom transformed the molecule from mutagen to nonmutagen (Figure 5a,b). Here, the predictions obtained from the model are in accordance with the experimental observation. Next, the fragment contributions using Grad-CAM score were used to analyze few of the incorrect predictions. In Figure 5c, the molecule on the right (PubChem CID: 115152) is an experimentally verified nonmutagen but was predicted to be an Ames mutagen. It was observed that a large part of the molecule (i.e., the main scaffold) was found to have a

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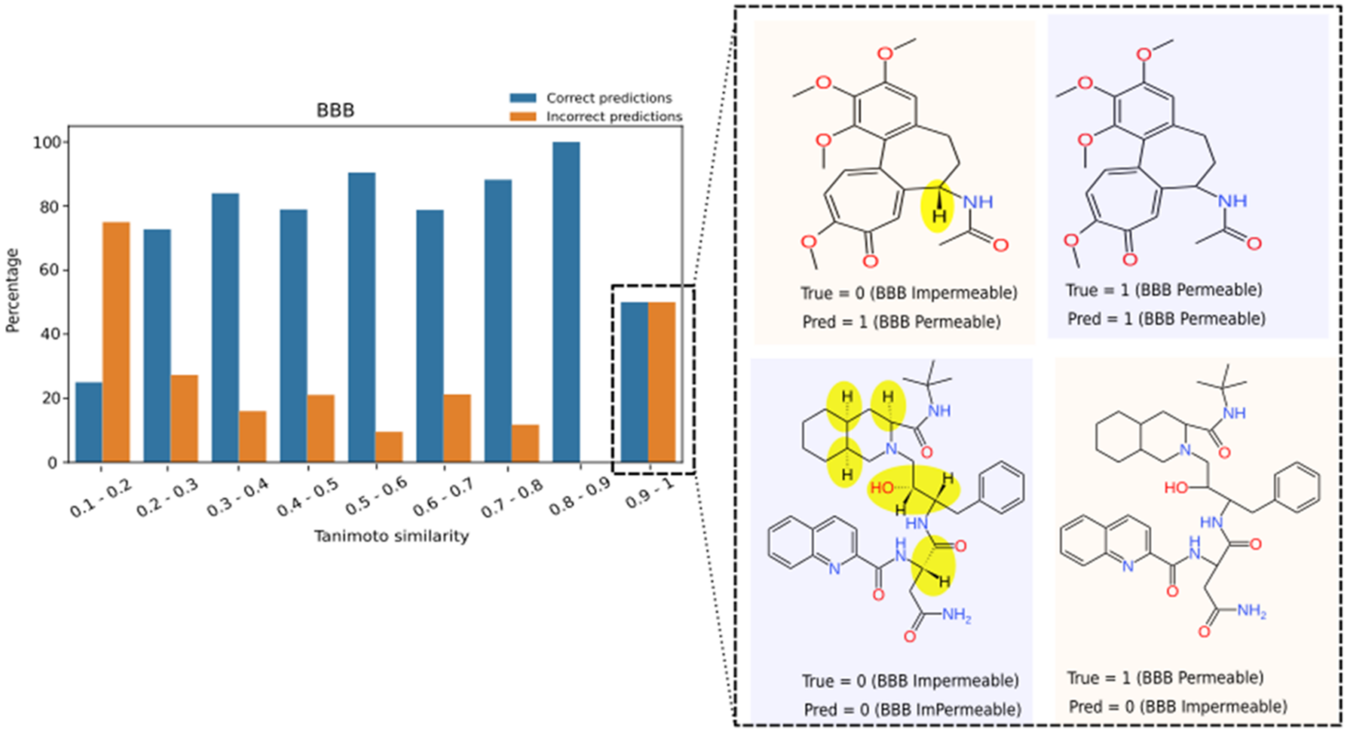

Figure 6. Performance of the BBB permeability model as a function of Tanimoto similarity shows that the model performance increases with the degree of similarity with molecules present in the training data set. The Tanimoto similarity of the test set is computed with respect to training set molecules. The correct and incorrect prediction in each bin is colored blue and orange, respectively. The inset shows the molecules that are identical but stereoisomers of the molecules present in the training data set and, as a result, wrongly predicted by the model.

positive Grad-CAM score. Also, the average Grad-CAM score of the fragment (hydroxyl group) responsible for MMP transformation was found to be positive in the mutagen data set. Similarly, in Figure 5d, the molecule on the left (PubChem CID: 55782) is a nonmutagen but was predicted to be a mutagen. In this case, Grad-CAM contribution of the MMP transformations (bromide and hydroxyl) was found to be positive toward the mutagen. As a result, both the molecules were predicted as the Ames mutagen by the model.

The model has also learned to differentiate subtle variations in fragments and assign different Grad-CAM contributions based on them. As a result, the benzene rings present in Figure 5 got different Grad-CAM scores. In Figure 5, mono- (indicated by the attachment points (*) in the fragment SMILES) and disubstituted benzenes have different Grad-Cam scores. Further, the nature of the substitution (polar/nonpolar) will also be captured by the neighborhood aggregation function of graph convolution networks, adding additional information necessary for the prediction.

However, the Grad-CAM approach, which offered a fragment-level explanation, cannot be applied for the regression tasks. The Grad-CAM method aims to highlight the significant input features associated with a specific class. In the case of classification, specific classes can be obtained. This is not possible for regression tasks. As a result, the Grad-CAM method cannot be directly employed to obtain explanations for regression tasks.

High Quality and Quantity of Data Can Improve the Model Prediction. Based on the performance of the model shown in Tables 2 and 3, it can be observed that while the model performance is comparable with the state-of-the-art, it has scope for improvement. Tables 2 and 3 also indicate the available data for each of the predictions. For example, the available data are as low as ∼278 for one of the properties. The quality of the data is also important. As described in Figure 4c,d, the explainable fragment method raised concern about the quality of the data in the case of MMPs related to BBB permeability.

At the same time, there must be enough diversity in the training data set. If the molecules from the test data are completely novel compared to the training data set, the model predictions can go wrong. This is particularly true for a low-data scenario. To further validate this point, the molecules from the test set were grouped into nine bins. By defining a bin value threshold, each bin contains correctly and incorrectly predicted properties of the small molecules from the test data set based on their degree of similarity with the training data set. The similarity was calculated based on the Tanimoto coefficient.[57]

Figure 6 depicts the normalized (based on frequency) values of correct and incorrect predictions for BBB permeability (for actual values, refer to Supporting Information, Figure S3). From Figure 6, it is evident that the performance of the model increases with increasing Tanimoto similarity to the training set. However, it was observed that two molecules with Tanimoto similarity 1 (identical) compared to the molecules in the training data set were incorrectly classified by the model. Upon further analysis, it was found that the molecules are stereoisomers and therefore belong to different classes (Figure 6, inset). This is one of the limitations of the current model. Therefore, the model performance would increase with the presence of diverse molecules in the training data set. Such a model would be more useful for predicting the properties of novel AI-generated molecules.

Overall, it was observed that the performance of models developed in the current study is better or at par with the models reported in the literature. The work further highlights the importance of fragment-level explainability, which can be useful

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for medicinal chemists to obtain explainability during lead modification and optimization. The high quality and quantity data are required to improve the model's prediction. At the same time, diversity in the training data is also required, which will help in predicting the property of novel molecules.

■ CONCLUSIONS

In summary, we have proposed a new fragmentation method named pBRICS for representing small molecules such that it captures a chemist's intuition. The performance of the model trained using pBRICS fragments is higher when compared to atom-based representation. Also, the multitask property prediction model trained separately for classification and regression tasks showed significant improvement over the corresponding multitask model using all 40 ADMET properties. The explanations obtained from the trained model are easily interpretable by a medicinal chemist and can be particularly useful for drug-like property optimization. The method was applied to predict the property of MMP pairs. While there was some success with the prediction of MMP for BBB permeability and Ames mutagenicity properties, the predictive performance is limited due to the amount of data that is available to train AI-based models. For a few cases, open-source data are questionable, as predicted by the explainable method. The proposed pBRICS method could guide the optimization of lead molecules, where the explanation obtained from deep learning models can be used to improve molecular properties.

■ ASSOCIATED CONTENT

Data Availability Statement

All input data are publicly available and a detailed description for the same is mentioned in the Materials and Methods section. The code used to generate results shown in this study is available from the corresponding author upon request for academic use only.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.3c00689.

Matched molecular pair analysis to evaluate BBB permeability and Ames mutagenicity using two models employed in this study (MT-Frag-mod and MT-Atom), along with an external model called SwissADME; performance (AuROC score) of proposed multitask models (MT-Frag and MT-Frag-mod) and their comparison with the MT-Atom (based on ADMETLab2.0) model, MT-BRICS, MT-RECAP, and MT-SynDiR on 30 classification data sets; the data set size includes train, validation, and test data; performance (Spearman Correlation) of the proposed multitask models (MT-Frag and MT-Frag-mod) and their comparison with the MT-Atom (based on ADMETLab2.0) model, MT-BRICS, MT-RECAP, and MT-SynDiR on 10 regression data sets; list of hyperparameters used for training the models in this work, including the hyperparameter names and their corresponding value ranges; results of the paired *t*-test conducted at a 0.05 level of significance on the models trained with the fragment-based graph input (MT-Frag and MT-Frag-mod) and atom- based graph (MT-Atom) input representation; comparison of the multitask models employed in study (MT-Frag-mod and MT-Atom) vs a multitask model reported in the literature (admetSAR2.0) on a subset of 17 common ADMET properties between the models; and results of the paired *t*-test at a 0.05 level of significance on MT-Frag-mod vs admetSAR2.0 (PDF)

Multitask data set of 40 ADMET properties (30 classification, 10 regression properties) (XLSX)

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Notes

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